

MIRRORED LOBES

WHY WE DON'T CALL THEM

The transition is still on the ~40° line, but the loops bulge the opposite way from the cal flaws. Here's what that means.

KEY IDEA

The angle says through-wall. The loops say it isn't behaving like a wall flaw. Don't size it on the flaw curve.

1 WHY THERE ARE LOOPS AT ALL

SOLID

- A differential channel shows coil A minus coil B.
- If the angle stayed exactly the same as the coils passed, you'd get a straight line, not loops.
- The loops exist because the angle shifts a little during the pass.

2 WHY THE ANGLE SHIFTS

REASONING

- Directly over the flaw, eddy currents reach it by the shortest path.
- Approaching or leaving, they reach it at an angle through more metal, so there's a little more lag.
- With a real wall flaw this drift always goes the same way. That's why all your cal flaws open the same way.

3 WHAT MIRRORED MEANS

- The drift runs the opposite way, so the signal doesn't behave like a flaw seen through the wall.
- Two things overlapping: a small flaw plus a dent, deposit, or support edge.
- A shape or fit change: a ding, dent, or probe wobble.
- Something outside the tube: a loose part, deposit, or support.

4 NOT THE SAME AS A REVERSED TRACE

- If the whole signal traces backward, check scan direction (push vs. pull) first.
- If everything in the tube is reversed, it's setup, not the signal.
- A lone reversed signal can mean added material, or the far end of a long flaw (it pairs with the start).



Same transition line, loops on opposite sides (illustrative).

CHECK IT

- › Compare with the ID groove and OD flaws on your cal standard
- › Mirrored on every channel, or only one?
- › Which way does the absolute channel swing?
- › Same spot on nearby tubes and previous data

IF THE ID GROOVE OPENS THIS WAY

- › Some setups may show ID signals like this. Ask your Level III how your procedure treats it.

QUICK TIPS

Your depth curve only works for signals that behave like the cal flaws.

'Reasoning' sections are physics-based. Confirm with your Level III.

Can't explain it? Flag it.